

DHWANI THAKKAR

Business Intelligence Engineer

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SUMMARY

- **Business Intelligence Engineer and Data Analyst with 4+ years of experience** specializing in big data analysis, data mining, statistical inference, A/B testing, machine learning, data visualization, and ETL pipeline development.
- Proficient in both SQL & NoSQL databases including PostgreSQL, SQL Server, MySQL, Oracle, MongoDB and skilled in writing stored procedures, triggers, and complex queries involving multiple joins, sub-queries, and window functions for report creation and data analysis.
- Good understanding on creating and optimizing data pipelines using Apache Airflow and AWS Glue, with a strong background in data warehousing solutions such as Snowflake.
- Excellent hands-on expertise of Tableau (Desktop, Reader, Server) and Power BI for building and deploying interactive reports and dashboards to drive business insights.
- Understanding of documentation & requirements gathering, including BRDs, FRDs, and SRS with good knowledge of Traceability Matrices to track requirements changes.
- Solid understanding of data integration, ETL processes, and performance optimization for large-scale data systems.
- Strong with Excel, including advanced functions like PivotTables, VLOOKUP, and Power Query, to extract and analyze data based on business needs.

SKILLS

Languages:	Python, R programming, SQL, SAS, MATLAB
Packages:	Pandas, NumPy, SciPy, Scikit-Learn, TensorFlow, Keras, Matplotlib, Seaborn
Databases:	MySQL, PostgreSQL, Oracle, MongoDB
Exploratory Data Analysis:	Data Warehousing, Data Wrangling, Data Modeling, Data Mining, Data Visualization
Data Science & Machine Learning:	Gen AI, Supervised Learning, Unsupervised Learning
Bigdata Technologies:	Hadoop, MapReduce, HDFS, Hive, Pig
Tools:	Tableau, Power BI, Looker, SharePoint, Jira, Visio, Lucidchart, Docker, Kubernetes, Terraform
Statistics:	Anova Testing, Bayesian Inference, Hypothesis Testing, A/B Testing
Clouds:	AWS (EC2, S3, R53, IAM, Redshift), Azure (Data Lake, Data factory), GCP
Others:	ETL, Airflow, Databricks, Risk Analysis, Snowflake, Decision Analysis, Financial Analysis, Documentation, Reporting, Traceability Matrices, Ad Hoc Analysis KPI Analysis
Methodologies:	SDLC, Agile, Scrum, Kanban, Waterfall
Operating Systems:	Windows, Mac OS, Linux

EXPERIENCE

BUSINESS INTELLIGENCE ENGINEER

| JAN 2025 – TILL DATE

Mphasis Inc., USA

- Led automation initiatives across inventory reporting processes by integrating Python scripts and VBA macros, reducing manual effort by 30% and ensuring consistent data validation.
- Utilized Python scripts and AWS Lambda functions to streamline repetitive tasks and enhance overall productivity.
- Ensured data integrity and consistency across datasets by implementing robust data validation frameworks using Python's Pandas and SQL and maintained a 95% accuracy rate.
- Performed advanced analysis and transformation on over 10,000 rows of data using Python (Pandas, NumPy) and SQL and applied data wrangling techniques to clean, join, and aggregate data, identifying key trends & actionable insights.
- Developed SQL-based reporting solutions for KPI tracking, providing insights into order accuracy, fulfillment rates, and supply chain performance to senior leadership.
- Spearheaded the migration to cloud-based analytics solutions, deploying AWS CloudWatch for real-time monitoring and Amazon S3 for optimized data storage.
- Designed and implemented automated Redshift dashboards using AWS QuickSight and VBA to visualize fulfillment KPIs, enabling real-time insights for business stakeholders and improving operational response time by 20%.
- Implemented cost-effective cloud storage strategies by leveraging Amazon S3 and optimizing data storage solutions.

- Conducted customer segmentation analysis using clustering algorithms and regression models, leading to a 12% increase in customer retention.
- Developed and optimized interactive Power BI dashboards using DAX, Power Query, and advanced visualizations to track and display marketing KPIs.
- Guided other analysts on advanced Excel-based reports, leveraging pivot tables, VLOOKUP, Power Query, and macros.

DATA ANALYST

| JUL 2020 – JUN 2024

LTI MINDTREE, India

- Automated data extraction and integration processes to reduce manual effort and improve data pipeline efficiency, achieving a 30% reduction in manual effort.
- Applied data wrangling techniques to restructure complex datasets, preparing them for advanced statistical modeling and visualization.
- Developed statistical models using Python (SciPy, Scikit-Learn) and R, including regression analysis and classification models, to predict outcomes and support strategic initiatives.
- Created data visualizations to communicate complex insights to non-technical stakeholders, enhancing decision-making processes across the organization.
- Provided decision support through risk and decision analysis, evaluating potential outcomes and recommending optimal strategies.
- Implemented best practices for database indexing, normalization, and performance tuning to enhance system efficiency.
- Led the design and execution of ETL processes using tools like Airflow and Glue, ensuring accurate and timely data movement between systems and data warehouses.
- Streamlined data pipeline processes, reducing data latency and improving the availability of real-time insights for business operations.
- Presented analytical findings and recommendations to stakeholders using PowerPoint and other communication tools, driving data-informed decisions.
- Utilized project management tools like JIRA and Confluence to track project progress, manage tasks, and ensure timely delivery of data analysis projects.
- Optimized and automated complex marketing workflows by design, implementing process maps by Visio & Lucidchart.
- Participated in Agile and Scrum processes, contributing to sprints and collaborating with team members to meet project milestones and deliver high-quality analytical outputs.
- Ensured data quality and integrity by implementing robust validation processes, maintaining accurate and reliable datasets for organizational use.

EDUCATION

Master of Science in Applied Data Analytics
Boston University, MA

GPA - 3.90

Master of Science in Data Science
Vellore Institute of Technology, India

GPA - 3.80
Aug 2020 – Jul 2022

Bachelor of Computer Applications and Computer Programming
Veer Narmad South Gujarat University, India

GPA - 3.80
Nov 2017 – Jul 2020

CERTIFICATION

AWS Cloud Practitioner

ACADEMIC PROJECT

Human Activity Recognition using ML and OpenCV:

- Developed a Human Activity Recognition (HAR) system leveraging accelerometer and gyroscope sensor data, achieving 91% classification accuracy across six distinct activities using machine learning algorithms such as SVM and Random Forest.
- Built a computer vision pipeline using OpenCV and Python to detect and count vehicles from video input, enabling real-time traffic density analysis and congestion prediction.
- Integrated data preprocessing, feature extraction, and model training techniques, demonstrating proficiency in sensor data analytics, supervised learning, and video processing automation.